Predicting Metal-Oxide Nanoparticle Cytotoxicity from Physicochemical Descriptors: A Comparative Machine Learning Study with Leakage-Aware Data Curation

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***Abstract*—** Engineered nanoparticles are increasingly used in drug delivery, where their small size and large specific surface area allow them to interact closely with cells and tissues. The same properties, however, can produce unwanted cytotoxicity, and experimental screening of every candidate formulation is slow and costly. This paper reports a data-driven framework that classifies metal-oxide nanoparticles as toxic or non-toxic from their physicochemical descriptors and exposure conditions. Working from a published dataset of 483 in-vitro experiments covering five metal oxides and eleven cell lines, we found that a naive treatment of the data produces badly inflated performance estimates for two distinct reasons. First, the cell-viability measurement from which the toxicity label was originally derived is present as a column in the dataset and acts as a direct label leak. Second, and less obviously, the file contains replicate experiments that become exactly identical once that column is removed; a conventional random split therefore places copies of the same experiment in both the training and test partitions. After correcting both issues the usable dataset reduces to 355 unique experiments, of which 19.2% are toxic. Five classifiers—logistic regression, a decision tree, k-nearest neighbours, a support vector machine and a random forest—were trained within a leakage-safe pipeline and compared. The random forest performed best, achieving 0.944 accuracy, 0.941 recall and an F1 score of 0.865 on a stratified held-out test set. We argue that recall, not accuracy, is the metric that matters for a screening tool: the k-nearest neighbours classifier attains 0.876 accuracy while detecting only 41.2% of toxic particles. Finally, a leave-one-material-out evaluation shows that performance degrades substantially on a material the model has not seen, with held-out ZnO accuracy falling to 0.771. We report this openly, since it indicates that models trained on the small number of oxides available in current public datasets generalise across exposure conditions far better than they generalise across chemistries.

***Keywords*—** nanotoxicity, nano-QSAR, machine learning, random forest, data leakage, class imbalance, drug delivery, cytotoxicity prediction

I. INTRODUCTION

Nanoparticles occupy an awkward position in biomedical engineering. The properties that make them attractive as drug carriers—dimensions of tens of nanometres, very high surface-area-to-volume ratios, and surfaces that can be functionalised to target particular tissues—are also the properties that allow them to cross membranes, accumulate in organs and generate reactive oxygen species. A carrier that delivers its payload efficiently is of little use if the carrier itself damages healthy cells.

Establishing whether a given formulation is safe is conventionally an experimental question, answered with in-vitro assays across a range of doses, exposure durations and cell lines. This is thorough but expensive, and it scales poorly: the design space spanned by material chemistry, particle size, surface charge and coating is far larger than any laboratory can enumerate. Computational screening is therefore attractive not as a replacement for assays but as a filter that decides which formulations are worth testing at all.

The established computational approach is nano-QSAR, in which a quantitative model relates structural or physicochemical descriptors of a nanomaterial to a measured biological response. Since the earliest such models the field has moved steadily from small hand-fitted regressions towards general-purpose machine learning classifiers trained on curated experimental collections. Our work sits in that second category.

This paper makes three contributions. We build and compare five standard classifiers for binary cytotoxicity prediction on a public metal-oxide dataset. We identify and correct two data-integrity problems in that dataset—a label leak and a replicate-duplication artefact—that between them inflate reported performance and, in our experiments, change which classifier appears to be the best. And we quantify how far the resulting model actually generalises by holding out entire materials rather than random rows, which we believe is the more honest test for this kind of data.

II. RELATED WORK

The foundational demonstration that nanoparticle toxicity can be predicted from computed descriptors rather than measured in every case is due to Puzyn and co-workers [1], who fitted a nano-QSAR model to the cytotoxicity of seventeen metal oxides against *E. coli* using a single quantum-chemical descriptor, the enthalpy of formation of the gaseous cation. The model was deliberately parsimonious and remains a reference point for the field.

Liu et al. [2] extended the idea from regression to classification, developing nanoSAR models that separate toxic from non-toxic metal-oxide nanoparticles and noting that classification is often the more defensible target when the underlying assay data are noisy. Their emphasis on classification over exact potency prediction motivated our own choice of a binary target.

As datasets grew, attention shifted towards their quality. Trinh et al. [3] curated metallic-nanoparticle datasets and explicitly assessed completeness and consistency before building nanoSAR classifiers, arguing that curation decisions materially affect the models that follow. Oh et al. [4] made a similar point at larger scale, assembling 307 publications on cadmium-containing quantum dots into a single meta-analysis and showing that surface properties and assay conditions, rather than core composition alone, drive the observed toxicity. Both papers informed the attention we paid to replicate structure in our own data, which turned out to matter more than any modelling choice we made.

On the modelling side, Furxhi et al. [5] compared a range of individual and ensemble classifiers for in-vitro nanoparticle toxicity and ranked them using the Copeland index, concluding that ensemble methods were generally preferable. Their later review [6] surveyed practices across the nanotoxicology machine learning literature and identified recurring weaknesses: small datasets, severe class imbalance, inconsistent validation protocols and an over-reliance on accuracy as the headline metric. Several of the design decisions described in Section III are direct responses to that critique.

The dataset used here was assembled by Subramanian and Palaniappan [7], who compiled metal-oxide cytotoxicity experiments with physicochemical and quantum-mechanical descriptors and fitted a deliberately parsimonious model to them. We use their data rather than their model, and our contribution relative to that work is the curation analysis and the material-wise generalisation assessment reported below.

III. MATERIALS AND METHODS

A. Dataset

We used the publicly available NanoTox metal-oxide nanoparticle dataset [7], distributed under an MIT licence. In its raw form it contains 483 in-vitro experiments described by 24 columns. Five metal oxides are represented—Al_2O_3, CuO, Fe_2O_3, TiO_2 and ZnO—tested against eleven cell lines, each labelled as cancerous or normal. Each record combines six experimentally measured or chosen quantities (core size, hydrodynamic size, surface charge, specific surface area, exposure time and dose) with thirteen material-level descriptors such as enthalpy of formation, band-edge energies, electronegativity, molecular weight and oxidation state. The dataset contained no missing values.

B. Label leakage

The dataset supplies both a continuous *viability* measurement and a binary *class* label. The latter was derived from the former by the original authors by thresholding. Retaining viability as a predictor therefore does not constitute prediction at all: it exposes a monotone transform of the target. Grouping the raw records by class confirms this, the two groups being separated almost perfectly along the viability axis. We excluded the column before any analysis, and enforced its exclusion with an automated test so that it could not be reintroduced by a later edit. We mention this explicitly because a model trained with the column present reaches near-perfect scores that are entirely spurious, and the trap is easy to fall into.

C. Replicate consolidation

A second and more subtle problem emerged during testing. Removing six exactly duplicated rows left 477 records, but a check for duplicates over the *feature* columns alone revealed a further 122 repeated rows. The explanation is that the source file records the same experimental condition (identical particle, dose, exposure time and cell line) several times, each with its own measured viability. While viability is present these rows are distinct; once it is removed they become identical.

The consequence for model evaluation is serious. A random train/test split distributes such replicates across both partitions, so the test set is no longer composed of unseen experiments and the reported score is optimistic. We therefore consolidated replicates to one row per unique experimental condition. Where replicates disagreed on the label—which occurred for 11 groups, reflecting genuine biological and measurement variability—we assigned the majority label, resolving exact ties in favour of the toxic class on the grounds that a false alarm is less costly than a missed hazard in a screening context. This reduced the dataset to 355 unique experiments, of which 68 (19.2%) are toxic.

D. Feature organisation

Of the nineteen numeric descriptors, thirteen are constant within a material: Al_2O_3 always carries the same molecular weight and oxidation state regardless of how the particle was prepared. We separated these material constants from the six quantities that genuinely vary between experiments. The distinction has a practical purpose—a user of the resulting tool need only supply six measurements and the material name, with the chemistry looked up automatically—but it also has an analytical consequence discussed in Section V. Cell type was retained as a single categorical variable and one-hot encoded. Material identity and cell line were deliberately excluded as predictors so that the model could not simply memorise them.

E. Classifiers and pipeline

Five classifiers were compared: logistic regression, a decision tree of maximum depth six, k-nearest neighbours with k = 5, a support vector machine with a radial basis kernel, and a random forest of 300 trees. Each was embedded in a single

pipeline together with the standard scaler and the one-hot encoder, so that scaling parameters are estimated on training folds only. Fitting a scaler on the full dataset before splitting is a common and easily overlooked source of optimistic bias, and the pipeline construction removes it structurally rather than by convention.

Given that only 19.2% of samples are toxic, all classifiers that support it were given balanced class weights. Without this correction the models converge on the degenerate solution of predicting the majority class everywhere, which attains 80.8% accuracy while identifying no toxic particles whatsoever.

F. Evaluation protocol

The data were split 75/25 with stratification on the target, giving 266 training and 89 test experiments (51 and 17 toxic respectively). We report accuracy, precision, recall, F1 and ROC-AUC on the held-out set, together with the mean and standard deviation of F1 under five-fold stratified cross-validation on the training portion. Model selection was performed on F1 rather than accuracy, for reasons developed in Section V. All random seeds were fixed, and the entire pipeline is reproducible from the raw file.

IV. RESULTS

A. Dataset characteristics

The curated dataset is strongly imbalanced and the imbalance is concentrated by material. Table I shows that toxicity is essentially confined to CuO and ZnO, with 38.9% and 33.5% of their respective records labelled toxic, while Al_2O_3 and Fe_2O_3 contribute no toxic records at all and TiO_2 contributes one. This is chemically unsurprising, since CuO and ZnO both dissolve appreciably at biological pH and release cytotoxic metal ions, whereas the other three oxides are comparatively inert.

Table I. Toxicity rate by material in the curated dataset.

Material	Experiments	Toxic	% toxic
CuO	18	7	38.9
ZnO	179	60	33.5
TiO_2	122	1	0.8
Al_2O_3	18	0	0.0
Fe_2O_3	18	0	0.0

B. Exploratory analysis

Comparing the toxic and non-toxic groups across the six controllable properties (Fig. 1) shows the expected dose-response behaviour: toxic records are associated with higher doses and longer exposure times. Toxic particles also tend to have larger core sizes and smaller specific surface areas in this dataset, though these trends are weaker and partly confounded with material identity. Hydrodynamic size showed almost no univariate relationship with the outcome, with a Pearson correlation of −0.079.

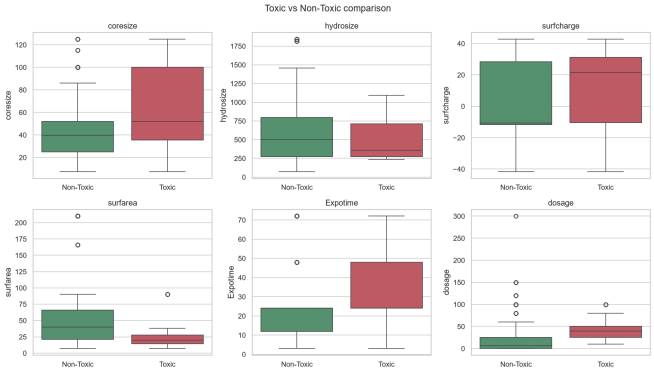


Fig. 1. Distribution of the six controllable properties for toxic and non-toxic experiments. Dose and exposure time separate the classes most clearly.

Ranking the descriptors by their linear correlation with the label places the material-level chemistry first: the metal-to-oxygen ratio (r = −0.415), enthalpy (−0.410), electronegativity per oxygen (+0.406) and oxidation state (−0.400) lead the list, with the strongest experimental variable, specific surface area, some way behind at −0.241. This is a direct consequence of the structure noted in Section III-D: because these descriptors are constant within a material, their correlation with the label is largely a restatement of the fact that CuO and ZnO are the toxic materials.

C. Classifier comparison

Table II reports held-out performance. The random forest is the strongest model on every metric except precision, reaching 0.944 accuracy, 0.941 recall, an F1 of 0.865 and a ROC-AUC of 0.946, and it is also the most stable under cross-validation (F1 0.880 ± 0.057).

Table II. Held-out test performance (89 experiments, 17 toxic). CV F1 is the mean and standard deviation over five stratified folds of the training set.

Model	Acc.	Prec.	Rec.	F1	AUC	CV F1
Random Forest	**0.944**	**0.800**	**0.941**	**0.865**	**0.946**	**0.880**
Decision Tree	0.921	0.778	0.824	0.800	0.904	0.831
Logistic Reg.	0.888	0.667	0.824	0.737	0.931	0.695
SVM (RBF)	0.865	0.609	0.824	0.700	0.928	0.665
KNN (k = 5)	0.876	0.875	0.412	0.560	0.836	0.527

The k-nearest neighbours result deserves comment. Its accuracy of 0.876 is higher than that of the support vector machine and only modestly below the decision tree, yet its recall is 0.412, meaning it fails to identify more than half of the toxic particles in the test set. It achieves its accuracy largely by agreeing with the majority class. A study reporting accuracy alone would rank this model respectably; by F1 it is clearly the weakest of the five. Fig. 2 shows the corresponding ROC curves.

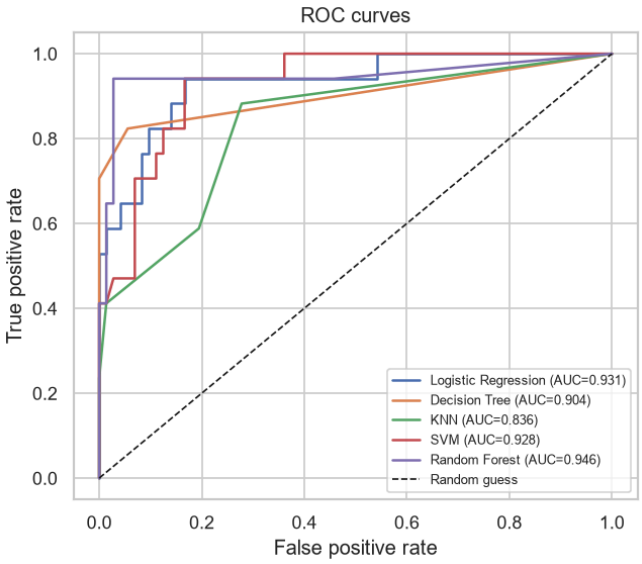


Fig. 2. ROC curves on the held-out test set. The random forest attains the highest area under the curve (0.946).

D. Feature importance

The random forest's impurity-based importances tell a different story from the correlation ranking. Dose dominates, accounting for 0.466 of the total importance, followed by exposure time at 0.100; the material descriptors follow at roughly 0.04–0.06 each (Fig. 3). The two views are not in conflict. Correlation is a linear, marginal measure and rewards descriptors that separate the materials, since material identity explains most of the variance in the label. The forest, by contrast, can split repeatedly on dose within a material and so exploits the dose-response relationship that the marginal correlation largely misses. For a practitioner the combined reading is the useful one: choice of material sets the baseline hazard, and dose and exposure duration modulate it.

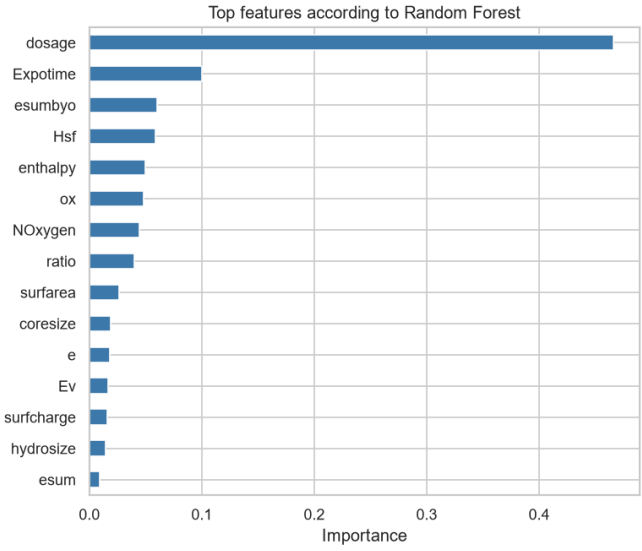


Fig. 3. Random forest feature importances. Dose and exposure time dominate, with material descriptors contributing at a lower level.

E. Leave-one-material-out evaluation

Because thirteen of the nineteen numeric descriptors are constant within a material, a random split allows the model to identify the material and apply a learned per-material prior. To measure how much of the reported performance depends on this, we retrained the selected model five times, each time holding out one material entirely (Table III).

Table III. Leave-one-material-out evaluation. Recall is undefined where the held-out material contains no toxic records.

Held out	n	Toxic	Acc.	Recall
Al_2O_3	18	0	1.000	–
Fe_2O_3	18	0	1.000	–
TiO_2	122	1	0.992	0.000
CuO	18	7	0.889	0.714
ZnO	179	60	0.771	0.717

The three folds that appear near-perfect are uninformative, since those materials contain essentially no toxic records and the majority-class prediction is almost always correct. The two informative folds are CuO and ZnO. Held-out ZnO accuracy falls to 0.771 from 0.944 on the random split, and recall to 0.717 from 0.941. The model does retain some genuine transferable signal—recall of roughly 0.71 on a chemistry it has never encountered is well above chance—but the degradation is substantial and would not be visible from the random-split figures alone.

V. DISCUSSION

A. Curation dominated modelling

The single change that most affected our results was not a modelling decision but a data-curation one. Before replicate consolidation, the decision tree appeared to be the best model with an F1 of 0.800, marginally ahead of the random forest. After consolidation the ordering reversed and the random forest led with 0.865. The decision tree had been partly memorising duplicated rows that appeared on both sides of the split, an advantage that disappeared once each experimental condition appeared exactly once. Held-out ZnO recall over the same change improved from 0.257 to 0.717, indicating that the uncurated model was not merely flattered by the split but genuinely worse. This is consistent with the emphasis that Trinh et al. [3] and Oh et al. [4] place on curation, and we would suggest that a duplicate check performed over feature columns only, after any label-derived column has been dropped, is worth making routine.

B. The choice of metric

With a 19.2% positive rate, accuracy is close to uninformative. The constant majority predictor scores 80.8%, and the KNN result in Table II shows that a real model can sit only a few points above that baseline while missing most of the hazards it exists to detect. In a screening application the two error types are not symmetric: a false positive costs an unnecessary assay, whereas a false negative allows a toxic formulation to advance. Recall and F1 are the appropriate criteria, and we would echo

the observation of Furxhi et al. [6] that accuracy-led reporting remains common in this literature and is misleading.

C. Limitations

The most important limitation is the narrow chemical coverage of the data. Five metal oxides, only two of which contribute meaningful numbers of toxic records, cannot support confident claims about nanoparticle toxicity in general, and the leave-one-material-out results in Table III should be read as an upper bound on what can be said about an unseen chemistry. The dataset is also small in absolute terms; with 17 toxic records in the test partition, individual metrics carry wide confidence intervals, which is why we report cross-validated F1 alongside the single-split figures.

Two further constraints follow from the data rather than the method. Toxicity here is a thresholded in-vitro viability outcome, and in-vitro viability is an imperfect proxy for in-vivo harm. And 11 replicate groups disagreed with themselves, which places a ceiling on the accuracy any model can legitimately reach on this data. Extending the work to polymeric and silica nanoparticles, applying resampling techniques such as SMOTE in place of class weighting, and using Shapley values to explain individual predictions rather than global importances are the natural next steps.

VI. CONCLUSION

We have presented a machine learning framework for classifying metal-oxide nanoparticles as toxic or non-toxic from physicochemical descriptors and exposure conditions. A random forest trained on 355 curated experiments reaches 0.944 accuracy and 0.941 recall on a held-out stratified test set, outperforming logistic regression, a decision tree, a support vector machine and k-nearest neighbours.

The more transferable findings, however, concern the data rather than the model. A label-derived column and an unnoticed replicate structure each inflate apparent performance, and correcting the second changed which classifier appeared best. A leave-one-material-out evaluation shows that performance on an unseen chemistry is markedly lower than a random split suggests. We would encourage both checks in future nanotoxicity modelling work, since neither is expensive to perform and both change the conclusions one is entitled to draw.

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